Sahaj Bhadja

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Summary

Software engineer with 2+ years building **performance-critical backend systems**, **low-latency data pipelines**, and **reliable concurrent workflows** using C++, Python, Java, SQL, and production CI/CD automation.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Operating Systems, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*

- Optimized **multi-tenant backend services** and indexed query paths for 500+ users, reducing latency from **1.2s to 350ms** while preserving deterministic behavior and stability across concurrent analytical workloads.
- Built **asynchronous FastAPI orchestration** for 5K+ monthly jobs with explicit state transitions, retry safeguards, and reconciliation checks, improving runtime reliability and reducing production failure propagation across dependent workflows.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*

- Implemented **high-throughput backend workflows** for 10K+ daily requests, coordinating concurrent model execution paths and API-level validations to sustain consistent response behavior under burst traffic conditions.
- Improved execution efficiency through **CI/CD automation** and controlled rollout patterns, reducing turnaround time by **60%** while maintaining correctness in fast-paced, production-facing feature releases.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*

- Engineered **microservice ETL workflows** with Python/Java OOP, AWS Glue, and S3, improving response times by **30%** while ensuring resilient data movement and predictable performance across distributed integration paths.
- Maintained **20+ automated unit/integration suites** using Jenkins, PyTest, and Mockito, boosting pipeline efficiency by **68%** and reducing defect carryover by **20%** across multi-environment deployments.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*

- Developed **Python evaluation pipelines** across 100K+ trajectory logs, using reproducible diagnostics and structured metrics to analyze model behavior under varied state transitions and constraint-driven decision scenarios.
- Produced implementation-focused technical reports that improved handoff quality between research and engineering teams, reducing integration ambiguity and accelerating iteration on production-bound algorithmic components.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*

- Built **real-time REST and WebSocket services** supporting eight workflows and 2M+ synchronized records, improving consistency of event propagation and robustness of stateful operations in live user-facing systems.
- Automated Docker/Jenkins deployment stages to reduce release time by **50%**, strengthening repeatability of build-test-release flows and enabling faster rollback-safe iterations in production-adjacent environments.

Skills

Languages: **C++**, **Python**, **Java**, JavaScript/TypeScript, SQL
Systems: low-latency optimization, concurrent/asynchronous workflows, OOP design patterns, memory-conscious coding
Backend: FastAPI, Flask, Node.js, REST APIs, microservices, PostgreSQL/MySQL
Infrastructure: Linux, Docker, Kubernetes, Jenkins, Git, CI/CD, observability-driven debugging

Projects

CPU Scheduling Simulator | *React, JavaScript, OS Algorithms*

- Implemented FCFS, SJF, SRT, RR, priority, and MLQ scheduling with interactive Gantt visualizations, reducing average turnaround by **25%** and improving debugging efficiency by **20%** across 1K+ simulations.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Built a **schema-aware SQL workspace** for 10+ databases with query validation and structured execution outputs, improving developer speed and reducing high-cost troubleshooting for complex analytical query workflows.

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB*

- Developed a **high-scale transaction platform** supporting 50K+ users and 1M+ operations, emphasizing predictable performance, secure integrations, and robust deployment patterns for latency-sensitive, real-time financial interactions.